

Stock Market Analysis

Technical Indicators and Risk Management

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Why I Chose This Topic

- Investing is an essential life skill, and I wanted to go beyond surface-level understanding
- Connects directly to my capstone project's theme of **financial wellness**
- Combines data analysis with financial literacy — skills that transfer to any career
- I wanted to understand what people actually mean when they talk about "reading charts"

How I Used AI to Learn

- **Claude** as my primary tutor — asked it to break down concepts step by step
- **ChatGPT** for second opinions when an explanation didn't click
- **Gemini / Perplexity** for quick fact-checks on finance terms and statistics
- I kept asking "**why**" and "**what's the catch**" until things actually made sense
- Cross-checked answers between models to make sure I was getting accurate info

Market Fundamentals

- **Exchanges** (NYSE, NASDAQ) match buyers and sellers through electronic order books
- **Market makers** provide liquidity by always quoting a buy and sell price — they profit from the spread
- **Order types** matter: market orders execute immediately at current price; limit orders only execute at your specified price or better
- Understanding these mechanics is the foundation everything else builds on

Fundamental vs. Technical Analysis

- **Fundamental analysis** reads the business: earnings, margins, debt, and valuation
- **Technical analysis** reads the chart: price, volume, and momentum signals
- Fundamental thinking plays out over months to years; technical over days to months
- Biggest takeaway from AI: **the strongest signals show up on both at the same time**

Fundamental Analysis — Reading the Business

- **Income Statement:** Revenue, expenses, and net income — is the company actually making money?
- **Balance Sheet:** Assets vs. liabilities — how healthy is the company's financial position?
- **Cash Flow Statement:** Where money comes from and goes — cash is what actually pays bills
- **P/E Ratio:** Price relative to earnings — a quick sanity check on whether a stock is overvalued
- **Debt-to-Equity:** How leveraged is the company — more debt means more risk

Moving Averages — Smoothing Out the Noise

- The **50-day** and **200-day** simple moving averages smooth out daily price noise
- **Golden cross**: the 50-day crossing above the 200-day — traditionally treated as bullish
- **Death cross**: the reverse — treated as a warning sign
- By themselves these are **lagging indicators** — useful for confirming a trend, not for timing entry

RSI — Relative Strength Index

- RSI compares recent gains to recent losses over a rolling window (usually 14 days)
- Above **70** is traditionally "overbought"; below **30** is "oversold"
- AI was blunt: RSI is a **confirmation tool**, not a buy or sell signal on its own
- Every shortcut I asked about had a real counter-example the model surfaced immediately

Risk-Adjusted Returns & Backtest Traps

- **Sharpe ratio** measures return relative to total volatility
- **Sortino ratio** only penalizes downside volatility — more realistic for most investors
- **Survivorship bias**: only looking at companies that still exist inflates historical returns
- **Look-ahead bias**: accidentally using future data in a backtest makes results useless
- A strategy that looks great on paper often falls apart once you account for real-world costs

Portfolio Diversification

- Spreading investments across sectors and asset classes reduces overall risk
- The goal isn't to eliminate risk — it's to **avoid catastrophic losses** from any single position
- **Asset allocation** (stocks vs. bonds vs. cash) matters more than picking individual stocks
- AI consistently reinforced: **position sizing and diversification are more important than stock picking**

My Interpretation — What Actually Stuck

- Successful investing combines **quantitative signals with disciplined risk management**
- Relying on gut feelings is how people lose money — indicators give you an **objective framework**
- No single indicator is a magic bullet; they work best **in combination** and with proper context
- The math behind moving averages, RSI, and Sharpe ratios is straightforward — the discipline to follow them is the hard part

How AI Changed My Understanding

- Before: I thought technical analysis was about "predicting" where a stock would go
- After: I understand it's about **measuring probabilities and managing risk**
- AI taught me to always ask for **trade-offs and limitations**, not just the "right" answer
- I verified concepts by asking multiple models and checking against real market data
- Without AI I wouldn't have covered this much ground — but the understanding is mine

Connection to My Capstone Project

- SmartSpend helps users manage personal finances with ML-powered budgets
- Stock market analysis reinforced the same core principle: **data-driven decisions beat intuition**
- Risk management in investing parallels budget management in personal finance
- Both domains reward the people who track numbers honestly instead of guessing

Questions?

Repository: github.com/Jolteer/ase485-learning-with-ai